Dimitrescu Robert 312CA - Partea I World Cup 2026 player value prediction

0. Pachete folosite pentru rulare cu venv (copy and paste), de asemenea nu am inclus player.csv in arhiva per cerintele temei. Kaggle Dataset:

adjustText==1.3.0 anyio==4.13.0 argon2-cffi==25.1.0 argon2-cffi-bindings==25.1.0 arrow==1.4.0 asttokens==3.0.1 async-lru==2.3.0 attrs==26.1.0 babel==2.18.0 beautifulsoup4==4.14.3 bleach==6.3.0 certifi==2026.5.20 cffi==2.0.0 charset-normalizer==3.4.7 comm==0.2.3 contourpy==1.3.3 cycler==0.12.1 debugpy==1.8.20 decorator==5.3.1 defusedxml==0.7.1 executing==2.2.1 fastjsonschema==2.21.2 fonttools==4.63.0 fqdn==1.5.1 h11==0.16.0 httpcore==1.0.9 httpx==0.28.1 idna==3.16 ipykernel==7.2.0 ipython==9.13.0 ipython_pygments_lexers==1.1.1 isoduration==20.11.0 jedi==0.20.0 Jinja2==3.1.6 joblib==1.5.3 json5==0.14.0 jsonpointer==3.1.1 jsonschema==4.26.0 jsonschema-specifications==2025.9.1 jupyter-events==0.12.1 jupyter-lsp==2.3.1 jupyter_client==8.8.0 jupyter_core==5.9.1 jupyter_server==2.18.2 jupyter_server_terminals==0.5.4 jupyterlab==4.5.7 jupyterlab_pygments==0.3.0 jupyterlab_server==2.28.0 kiwisolver==1.5.0 lark==1.3.1 lxml==6.1.1 MarkupSafe==3.0.3 matplotlib==3.10.9 matplotlib-inline==0.2.2 mistune==3.2.1 nbclient==0.10.4 nbconvert==7.17.1 nbformat==5.10.4 nest-asyncio==1.6.0 notebook==7.5.6 notebook_shim==0.2.4 numpy==2.4.6 packaging==26.2 pandas==3.0.3 pandocfilters==1.5.1 parso==0.8.7 pexpect==4.9.0 pillow==12.2.0 platformdirs==4.9.6 prometheus_client==0.25.0 prompt_toolkit==3.0.52 psutil==7.2.2 ptyprocess==0.7.0 pure_eval==0.2.3 pycparser==3.0 Pygments==2.20.0 pyparsing==3.3.2 python-dateutil==2.9.0.post0 python-json-logger==4.1.0 PyYAML==6.0.3 pyzmq==27.1.0 referencing==0.37.0 requests==2.34.2 rfc3339-validator==0.1.4 rfc3986-validator==0.1.1 rfc3987-syntax==1.1.0 rpds-py==0.30.0 scikit-learn==1.8.0 scipy==1.17.1 seaborn==0.13.2 Send2Trash==2.1.0 setuptools==82.0.1 six==1.17.0 soupsieve==2.8.4 stack-data==0.6.3 tabulate==0.10.0 terminado==0.18.1 threadpoolctl==3.6.0 tinycss2==1.4.0 tornado==6.5.6 traitlets==5.15.0 typing_extensions==4.15.0 tzdata==2026.2 uri-template==1.3.0 urllib3==2.7.0 wcwidth==0.7.0 webcolors==25.10.0 webencodings==0.5.1 websocket-client==1.9.0

1. Tipul problemei

Regresie — modelul prezice care este valoarea curenta a jucatorilor care vor juca la cupa mondiala din 2026 (market value).

2. Construcția dataset-ului

Am preluat date din doua surse, prima incadrandu se in " Web scraping / API-uri publice " si cea de a doua in " Extindere sau combinare de dataset-uri existente":

Wikipedia (2026 FIFA World Cup squads) — folosing pandas.read_html extragem coloanele:

- official appearances a player has made for the national team
- for what club they play
- age
- name
- position

Kaggle Player Scores (Transfermarkt) cu download pe sistem local, facem match numai pe jucatorii extrasi din wikipedia deja si adaugam pentru fiecare:

- player country
- current market value
- preak market value
- boolean check if a player is in his prime

3. Structura subseturilor

- Train: 740 linii
- Test: 247 linii
- Avem 10 coloane (features) + inca o coloana pe care o folosim drept target

Coloane	Tipuri
Player	object
Position	category
Age	int64
Appereances	int64
Goals	int64

Club	category
Club League	category
Country	category
Current MV	int64
Peak MV	int64
Apex MV	bool

	Player	Position	Age	Apperances	Goals	Club	Club League	Country	Current MV	Peak MV	Apex MV
0	Vladimír Coufal	DF	33	61	2	TSG Hoffenheim	L1	Czech Republic	2700000	12000000	False
1	David Zima	DF	25	24	1	Slavia Prague	TS1	Czech Republic	7000000	7500000	False
2	Jaroslav Zelený	DF	33	21	0	Sparta Prague	TS1	Czech Republic	600000	1500000	False
3	David Jurásek	DF	25	16	1	Slavia Prague	TS1	Czech Republic	5000000	8000000	False
4	Tomáš Souček	MF	31	89	17	West Ham United	GB1	Czech Republic	12000000	45000000	False
5	Vladimír Darida	MF	35	78	8	Hradec Králové	TS1	Czech Republic	375000	10000000	False
6	Michal Sadílek	MF	27	33	1	Slavia Prague	TS1	Czech Republic	8000000	8000000	True
7	Pavel Bucha	MF	28	0	0	FC Cincinnati	MLS1	Czech Republic	4000000	4000000	True
8	Alexandr Sojka	MF	23	0	0	Viktoria Plzeň	TS1	Czech Republic	2300000	2300000	True
9	Patrik Schick	FW	30	52	25	Bayer Leverkusen	L1	Czech Republic	20000000	50000000	False
...											
983	Carlos Harvey	MF	26	25	2	Minnesota United FC	MLS1	Panama	600000	600000	True
984	Azarias Londoño	MF	24	10	0	Universidad Católica	Unkown	Panama	1200000	1200000	True
985	Ismael Díaz	FW	29	54	17	León	MEX1	Panama	1800000	2000000	False
986	Tomás Rodríguez	FW	27	11	3	Saprissa	Unkown	Panama	1000000	1000000	True

4. Salvare CSV

Am salvat subseturile sub forma de CSV in test.csv si train.csv

5. Contributia proprie

Informatiile despre fiecare jucator din cupa mondiala impreuna cu features-ruile pe care le-am adaugat nu pot fi gasite pe net atltundeva. Ce am adaugat:

- am curatat valorile numerice din age sa fie in format de numar si nu string
- pentru coloana 'Player' am lasat doar nuemele jucatorului
- am redefinit tipurile tuturor obiectelor
- am normalizat numele unor tari care nu mai exista pentru jucatori mai in varsta
- am adaugat o coloana noua care verifica daca un jucator e in prime-ul lui
- am umplut valorile lipsa pentru jucatori fara club

- am eliminat valorile lipsa din restul coloanelor

6. Analiza exploratorie a datelor (EDA) complex

Toate observatiile mele despre date se afla in notebook. Am decis sa nu mai dau paste aici deoarece as repeta markdown-ul din notebook si ar fi clutered.

6.a Analiza valorilor lipsa inainte de inlocuire

	Number missing	Procent missing
Player	0	0
Position	0	0
Age	0	0
Appereances	0	0
Goals	0	0
Club	0	0
Club League	398	31.1912
Country	273	21.395
Current MV	289	22.6489
Peak MV	289	22.6489

6.b Statistici descriptive

Test set data dimension: (247, 10) Training set data dimension: (740, 10)

Part I : X numerical values for training and testing comparison:

X_train numerical statistics:

	Age	Appereances	Goals	Peak MV
count	740	740	740	740
mean	26.9878	28.5635	3.47838	2.07927e+07
std	4.11389	27.9076	7.46654	2.77643e+07
min	17	0	0	25000
25%	24	6	0	2.5e+06
50%	27	20.5	1	1e+07
75%	30	44	3	3e+07
max	43	152	79	2e+08

X_test numerical statistics:

	Age	Appereances	Goals	Peak MV
count	247	247	247	247
mean	27	28.3239	3.49798	2.08099e+07
std	4.04949	26.3985	8.39564	2.82257e+07
min	19	0	0	150000
25%	24	7.5	0	2.5e+06
50%	27	23	1	9e+06
75%	30	38	4	3e+07
max	38	130	89	1.5e+08

Part II : X non_numerical values for training and testing comparison:

X_train non_numerical statistics:

	Position	Club	Club League	Country	Apex MV
count	740	740	740	740	740
unique	4	327	32	45	2
top	DF	Crystal Palace	GB1	Argentina	0
freq	250	12	121	43	430

X_test non_numerical statistics:

	Position	Club	Club League	Country	Apex MV
count	247	247	247	247	247
unique	4	173	26	44	2
top	MF	Al-Nassr	GB1	Argentina	0
freq	84	5	38	12	148

Part III : Y values for training and testing comparison:

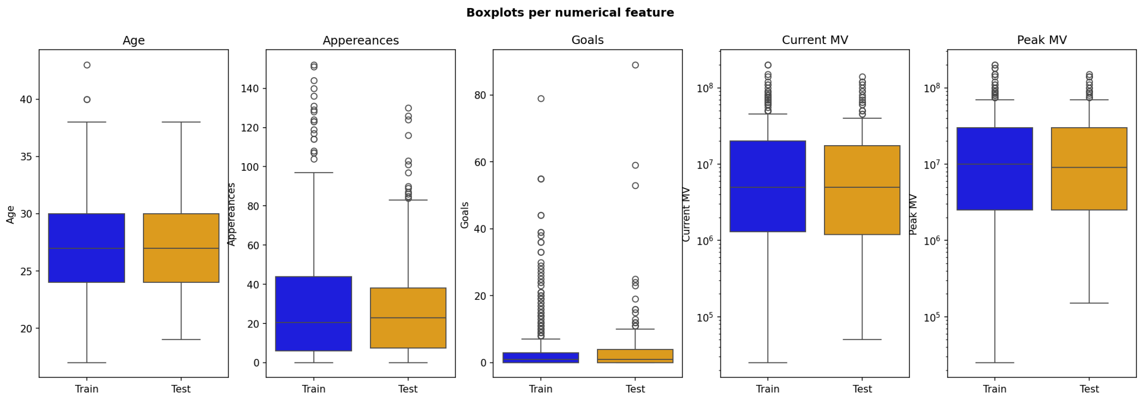
y_train statistics:

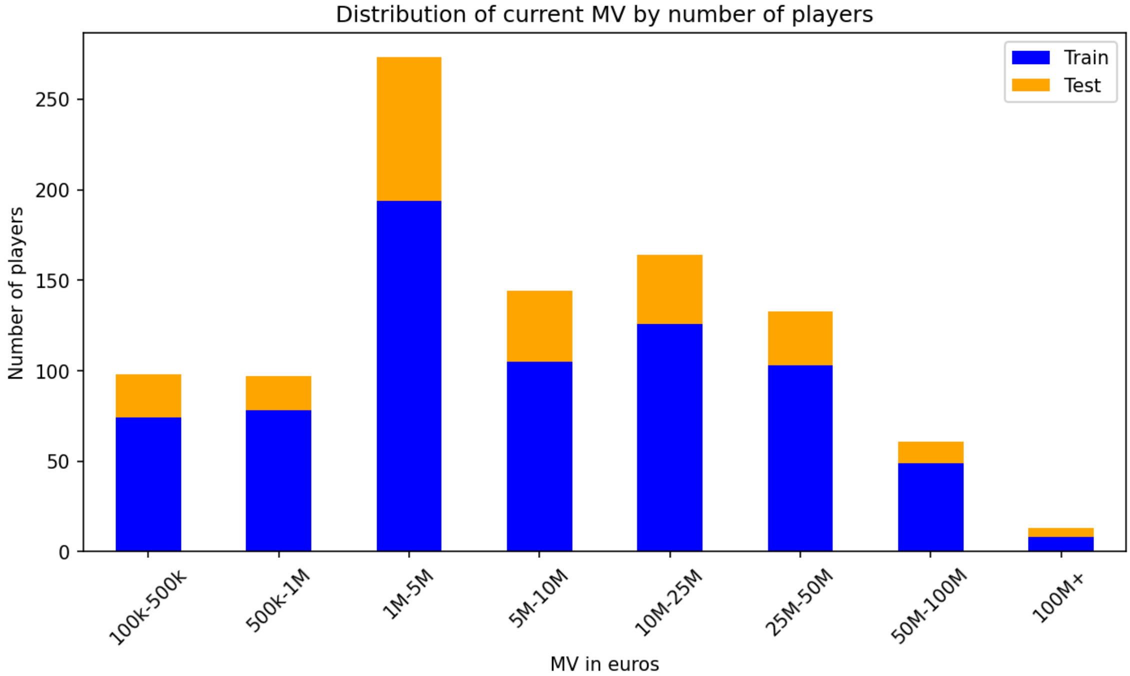
	Current MV
count	740
mean	14776182
std	22642161
min	25000
25%	1300000
50%	5000000
75%	20000000
max	200000000

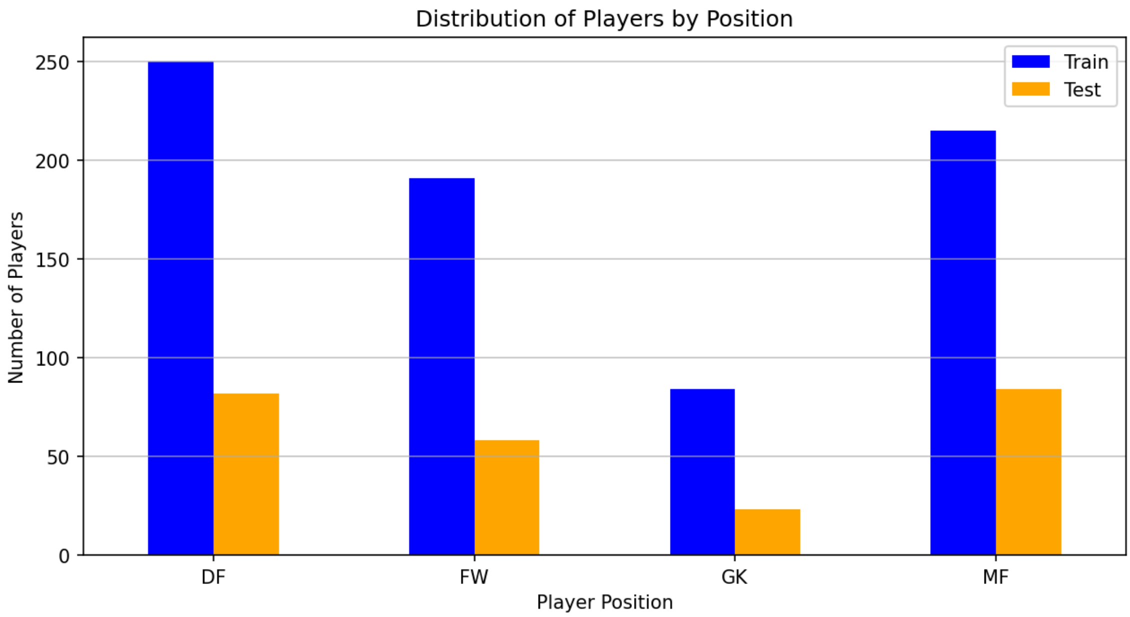
y_test statistics:

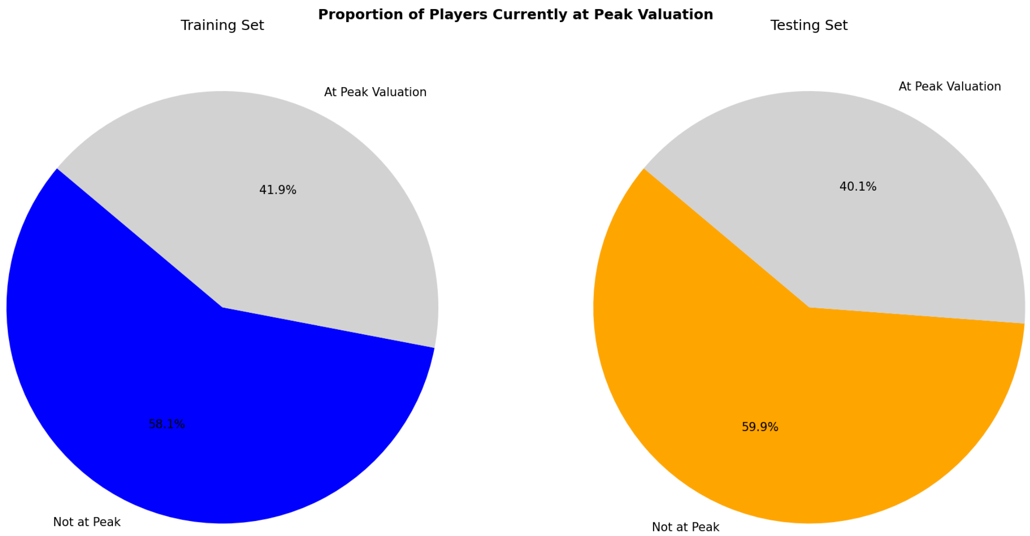
	Current MV
count	247
mean	14037753
std	22503640
min	50000
25%	1200000
50%	5000000
75%	17500000
max	140000000

6.c Analiza distributiei variabilelor

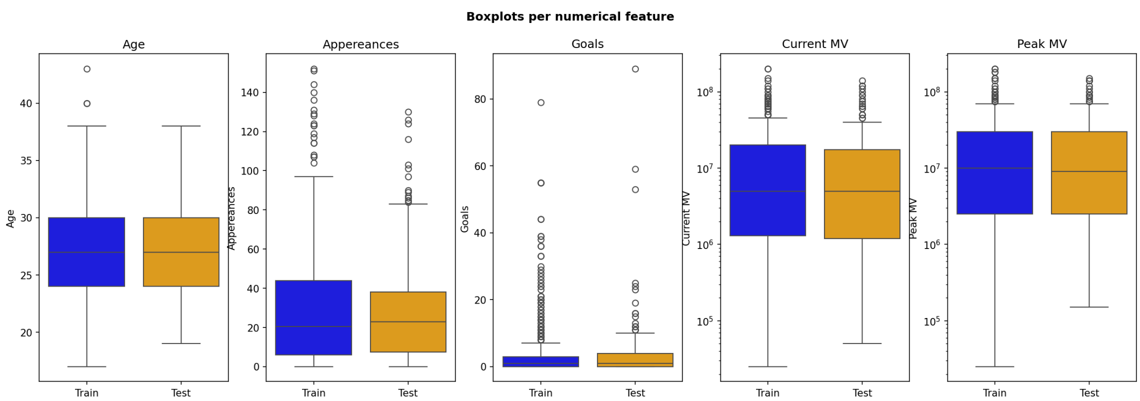

Boxplots per numerical feature
Age
Appereances
Goals
Current MV
Peak MV
Train
Test


Distribution of current MV by number of players
Train
Test
Number of players
MV in euros
100k-500k
500k-1M
1M-5M
5M-10M
10M-25M
25M-50M
50M-100M
100M+

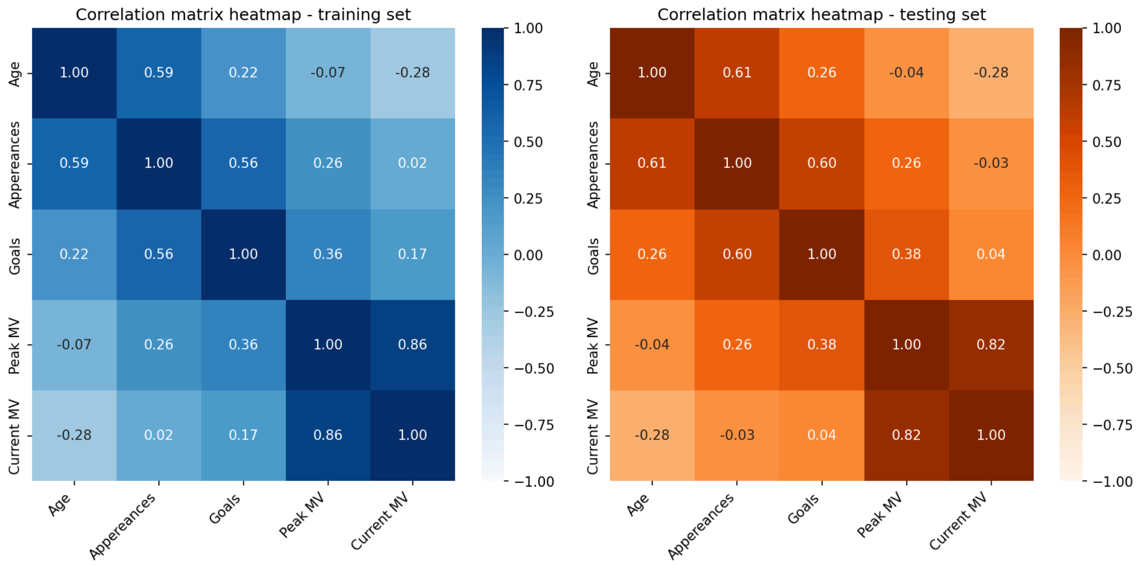




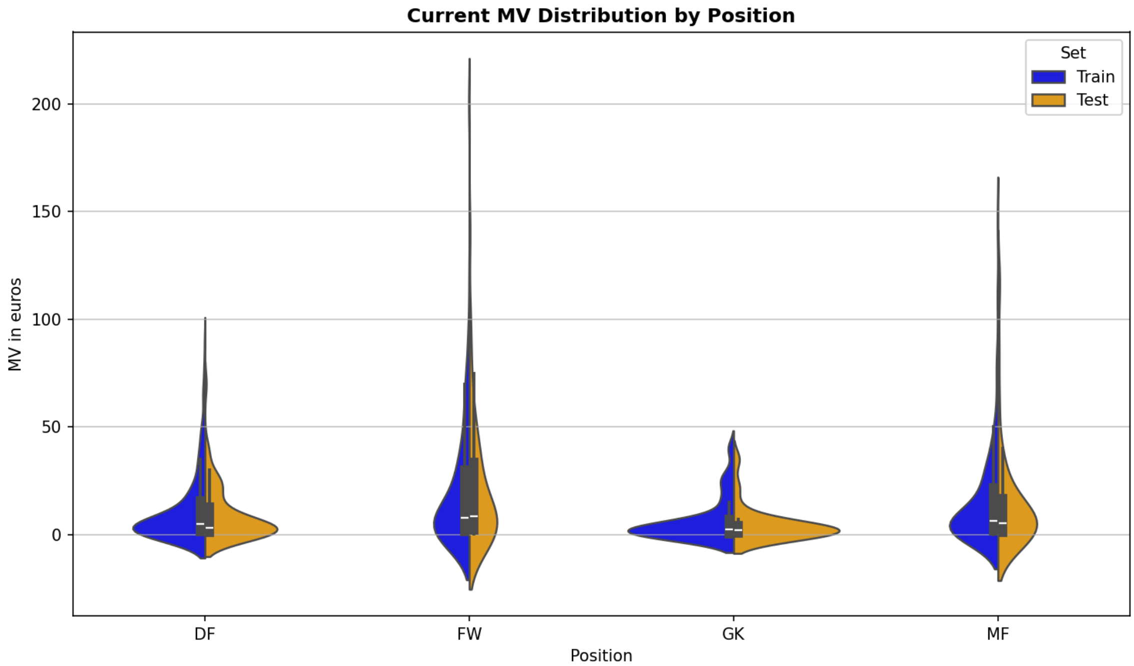
6.d Detectarea outlierilor

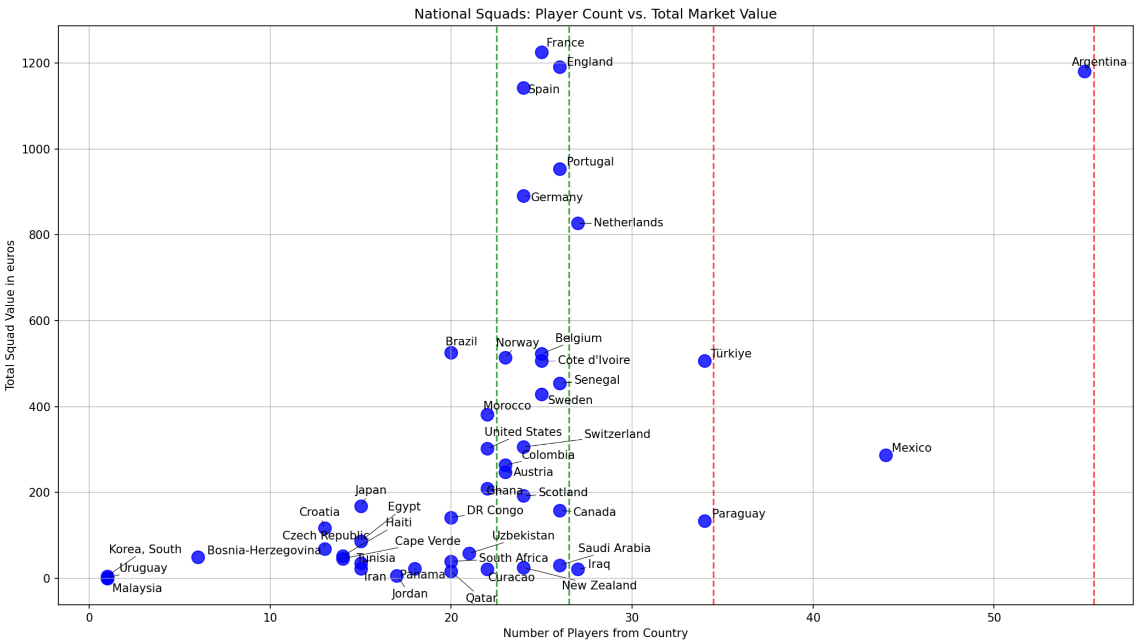


6.e Analiza corelatiilor



6.f Analiza relatiilor cu variabila țintă





6.g Comentarii si interpretări personale

Avem markdown detailat pentru fiecare celula si explicat mai in detaliu decat in README + comments in cod.

7. Antrenarea și evaluarea modelului

Preprocesare: one-hot encoding pentru Position, Club, Club League, Country. Avem problema ca modedelul nu poate privii obiecte/string uri cum le privim noi asa ca le transformam valorile in numere cu care poate lucra mult mai usor. Restul valorilor sunt numere deci nu avem probleme.

Model ales: RandomForestRegressor (n_estimators=200, random_state=42). Am ales acest model deoarece fata de regresia lineara simpla modelul se descurca foarte bine cu outliers (precum jucatorul nostru de 200M).

Rezultate pe setul de test: RMSE: 6,086,166 EUR MAE: 2,503,894 EUR R^2: 0.9266

Interpretare:

- R^2 ne arata ca modelul are o rata de aproximare de 92% pentru valoarea jucatorilor din piata si este un rezultat suficient de bun considerand setul nostru de date.
- MAE indica devierea in medie da la valoarea actuala cu 2.5 milioane de euro fata de valoarea reala.

- RMSE indica faptul ca pentru jucatorii foarte valorosi care sunt definite outliers pretul lor este subestimat. Accest lucru se intampla deoarce majoritatea jucatorilor nostrii se incadreaza intre 0 - 25 M de euro.
- Cel mai important factor de predicite a fost PEAK_MV deoarce valorile jucatorilor nu tind sa abata prea mult de la valoarea lor maxima inregistrata.